

Simulated Mars Swarm Exploration and Mapping

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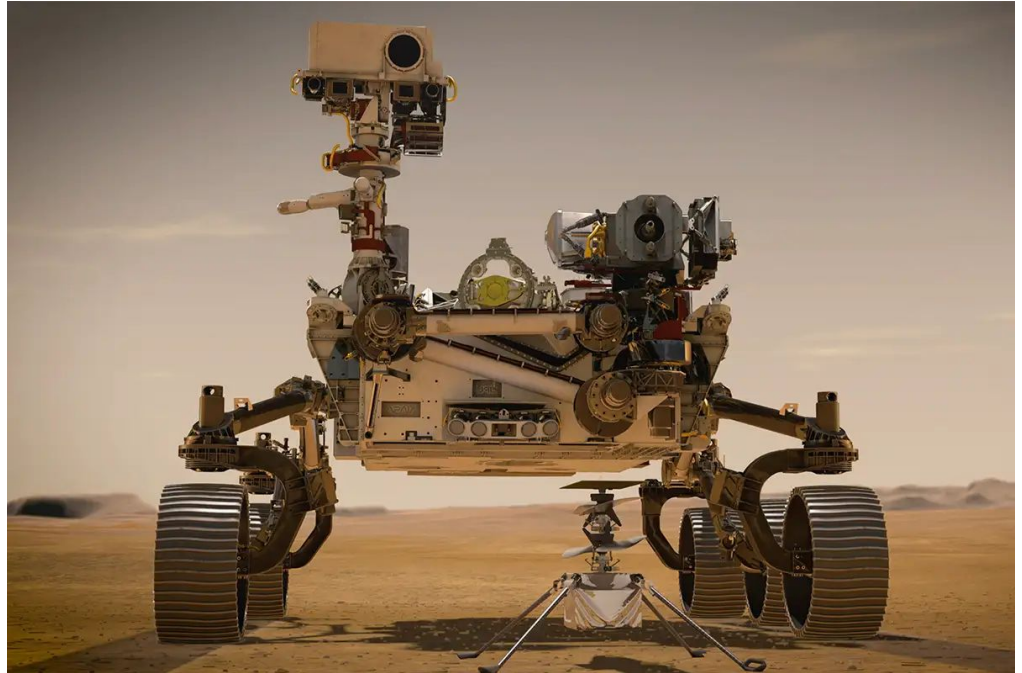
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NASA Space Apps Challenge

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Problem

Rover missions are
limited in time and
efficiency

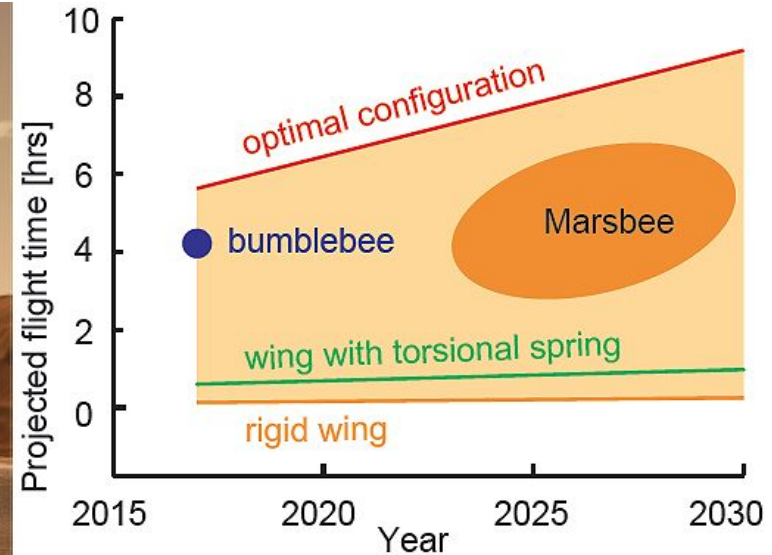
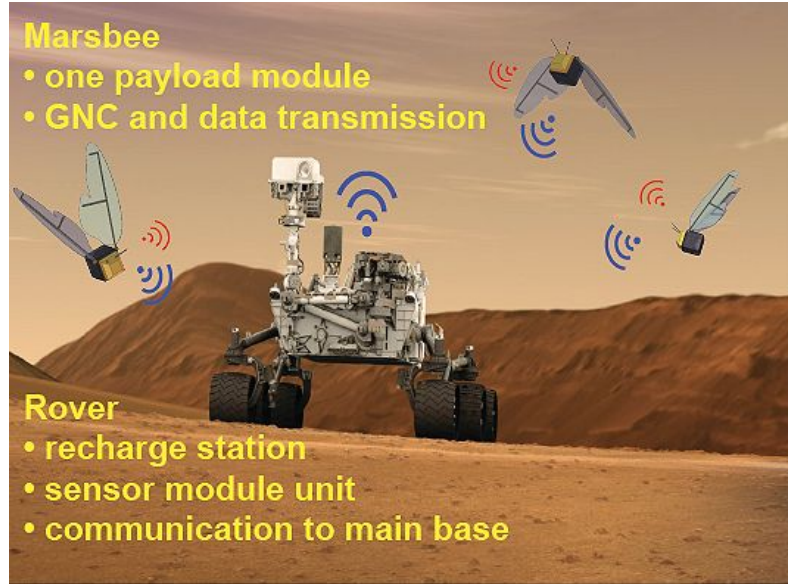


Current Solution

Ingenuity helps with this



Marsbee concept



Comparison

Attribute	Ingenuity Helicopter	Marsbee Concept
Volume	~31 L	~0.1 L*
Mass	1.8 kg	~0.028-0.061 kg
Power	350 W	Not available
Range	~0.3 km**	2.5km
Flight time	Up to 90 seconds per flight	53-114 minutes

*based on bumblebee size

**based on test flight parameters

<https://www.nasa.gov/general/marsbee-swarm-of-flapping-wing-flyers-for-enhanced-mars-exploration/>

Communication - Flying ad-hoc network (FANET)

Objectives:

1. **Enable Effective Inter-Swarm Communication:** Allow drones to share critical information in real time.
2. **Facilitate Data Gathering:** Ensure timely transmission of environmental data back to the rover.
3. **Optimize Resource Usage:** Minimize energy consumption and processing load for each drone.

Motivation: Synchronization in a swarm is crucial for mission success. We explored two main approaches:

- **Sensory Detection:** Each drone uses onboard sensors to detect obstacles. While effective, this method is CPU-intensive and drains battery life due to heavy computer vision processing.
- **Telecommunication Networks:** By leveraging coordinate calculations, drones can avoid collisions more efficiently, requiring less computational power.

On Mars, we have two primary communication options: UAV direct communication and ad hoc networks. Given the constraints of the Martian environment, ad hoc networks present a more scalable solution.

Communication Methods: We evaluated two communication strategies:

1. **UAV Direct Communication:** This method offers fault tolerance but leads to increased bandwidth usage as the swarm size grows, resulting in higher costs and energy consumption.
2. **Ad Hoc Networks:** Only specialized drones communicate directly with the rover, reducing overall bandwidth requirements and costs. This approach maintains constant bandwidth regardless of swarm size and allows for internal task distribution, reducing the load on the rover.

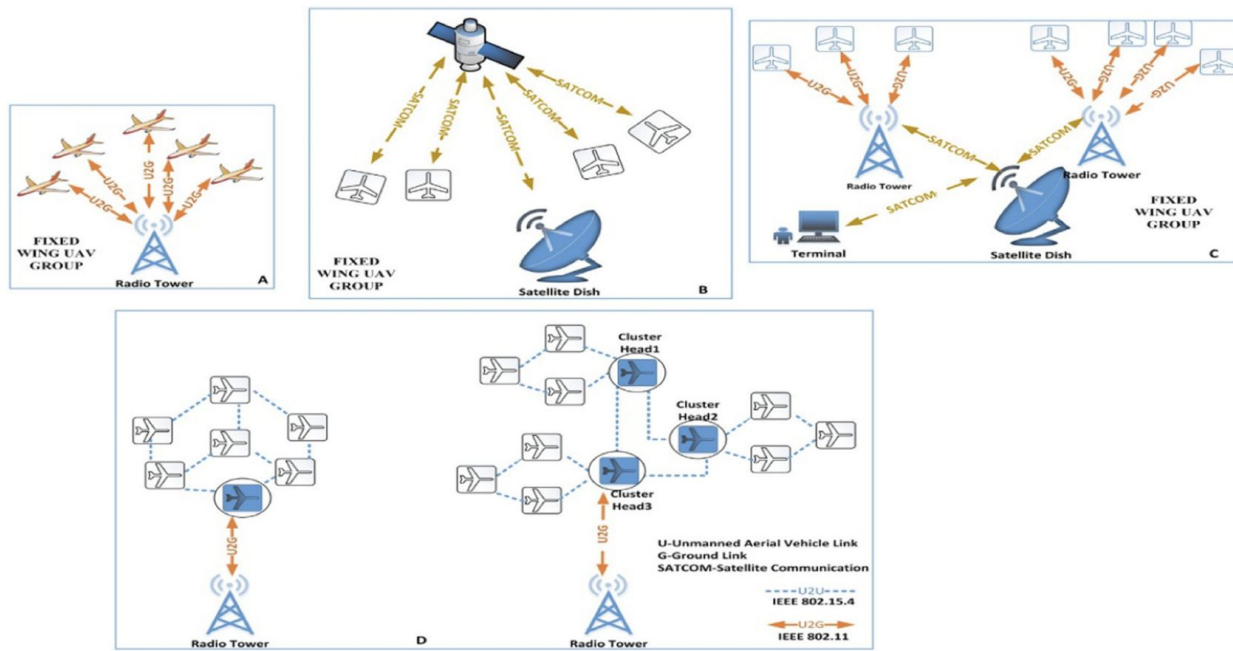


Fig. 3. Different Communication Nodes [50].

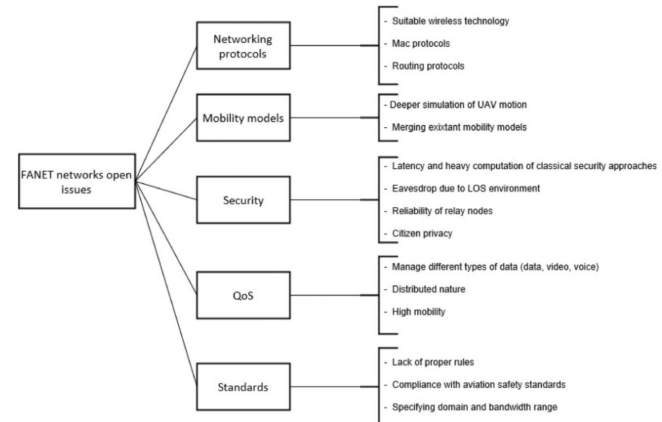
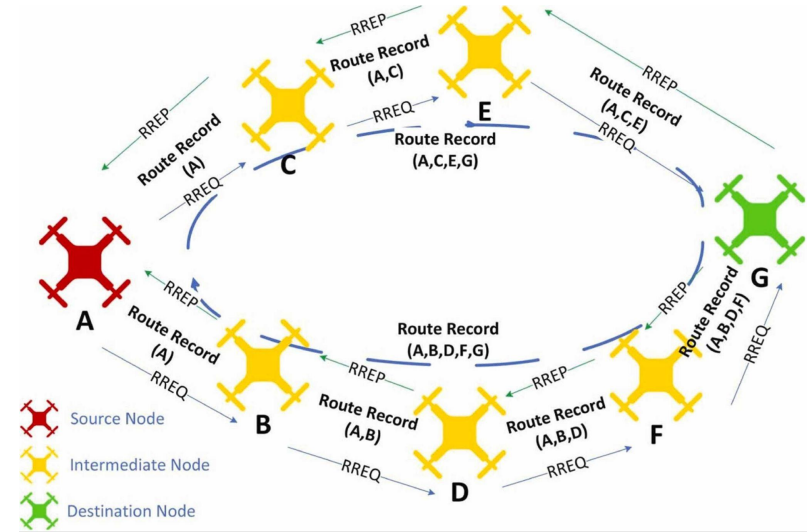
Architecture: In our architecture, each drone is represented as a node in a network graph. We will equip drones with LIDAR sensors for terrain mapping and cameras for image capture. This data will be relayed efficiently back to the rover, leveraging short-range, high-speed communication like 4G/5G.

Data Management: To handle large data sets from LIDAR scans and images:

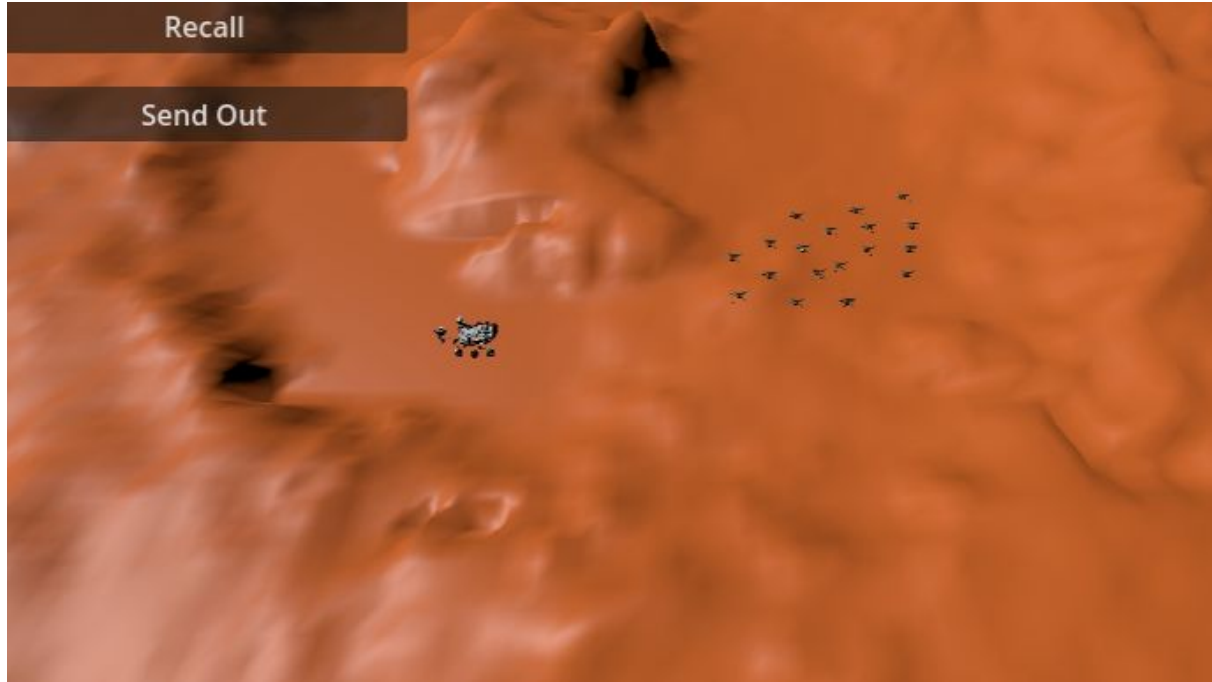
- implement file compression techniques to minimize transmission times.
- Utilize machine learning for data reconstruction and interpolation, allowing drones to process data onboard before sending it.

Data Transmission Protocols: Drones will employ a handshake method to confirm successful data packet transmission. If a drone loses connection, the swarm can adapt by either moving on or re-measuring from alternate locations. We will use established protocols like CSMA/CA to manage data interference effectively.

Challenges and Solutions: One challenge we face is potential network partitioning if ground control fails. However, our architecture allows for localized decision-making within the swarm, ensuring continuity of operations even when communication with the rover is temporarily lost.



Demo game



<https://ericc1231.itch.io/mars-bees-swarm-simulation>

Impact

- Mapping and exploration
 - Larger area mapped (kilometers or flight range, hours of flight time)
 - Extended pathfinding, many perspectives, 3D scans
- Resilience
 - No single point of failure
 - Can lose many drones -> still maintain capability
- Rover preservation
 - Mitigate damage to the rover
- Increase rover's autonomous capability

Website with our
write up:



<https://marsbeeswarmsimulation.netlify.app/>